

ARYA SALWAN



SENIOR COMPUTER
ENGINEERING STUDENT

CONTACT INFORMATION

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EDUCATION

Memorial University of Newfoundland

Bachelors of Computer Engineering

Co-op Program, (2021-2026)

AWARDS & ACHIEVEMENTS

- MUN International Undergraduate Student Scholarship (\$12,000) - 2021
- Verafin Scholarship (\$3,000)-2022

SKILLS AND ABILITIES

- **Programming:** Python, R, C#, Java, JavaScript, php, HTML, CSS, GoLang, SQL, Assembly, C/C++, VHDL and Swift.
- **Libraries:** PyQt, Pandas, Numpy, SciPy, Tensorflow, PyTorch, SciKit, VTK, Gdist, Jupyter, yfinance, ta
- **Frameworks:** NVIDIA OptiX, CUDA, React.js, Node.js, Angular, ASP.NET, .NET MAUI, Django, Nuxt.js & Flask,
- **DBMS:** MongoDB, PostgreSQL and MySQL.
- **OS:** FreeRTOS, Linux, MacOS, Android, IOS, Windows, QNX, FreeBSD, Embedded Linux & ROS.
- **DevOps:** Git, Jenkins, Kubernetes, Docker, Ansible, GCP, AWS and Azure.
- **Soft Skills:** Excellent communication skills, skilled debater and public speaker. Strong collaboration and leadership skills with a proven ability to collaborate effectively within a team.
- **Network Engineering:** Proficient in RFC, TCP/IP, DHCP, DNS, UDP, TCP, & UART protocols.

RELEVANT PROJECTS

MegaBot:

- Autonomous swarm robotics platform featuring a custom-designed integrated PCB (Pi CM5 & ESP32), high-precision localization, and an on-prem navigation for real-time autonomous research.

Hellbent Blade Game:

- A C++ based platformer game using ECS architecture, featuring a level editor, raycast based enemy AI, animations and inventories. Developed using SFML, & ImGui.

MUN Class Schedule Exporter:

- Chrome extension that exports MUN class schedule to electronic calendar.

BriskPDF Editor:

- Cross-platform PDF editing tool that allows users to Merge, Split, Insert & Password protect PDF files & View PDF files in DarkMode.

WORK EXPERIENCE

Application Software Engineer

Nasdaq Verafin, St John's

(2025 September - December)

- Engineered high-performance, interactive UI components for the Verafin platform using React.js, including a custom dynamic theming engine that delivered a seamless Dark Mode experience.
- Utilized Gatling framework to develop a suite of API performance tests designed to automatically detect any vulnerabilities and performance lag in the backend.
- Optimized data retrieval by writing complex SQL queries and analyzing backend API endpoints to resolve critical performance bottlenecks.

Full Stack Software Developer

Angler Solutions, St John's

(2025 January - April)

- Independently developed web-based simulation and data visualization platform for a solar energy using FastAPI, PostgreSQL, and Vue/Nuxt.js.
- Built RESTful APIs with FastAPI, implemented API key-based auth, and used asyncio to reduce response time by 75% under high-concurrency loads.
- Developed modular frontend with Vue.js/Nuxt.js using ECharts for making interactive graphs and heat maps; created landing page and UI elements based on Figma designs.
- Designed scalable databases using SQLAlchemy and Alembic to store scenario data while resolving performance issues.

Holographic Simulator Developer

Visual and Analytic Computing Lab, MUN

(2025 May - August)

- Upgraded the Lightfield simulator to CUDA 12.8, leveraging the latest parallel computing primitives and memory management features to enhance simulation performance.
- Traced and modernized complex OptiX ray-tracing pipelines within a C++ environment to ensure compatibility with modern NVIDIA architectures and updated SDK requirements.
- Managed the interaction between spatial data structures and real-time rendering loops, specifically focusing on the high-bandwidth requirements of Lightfield/Holographic displays.

Visual Computing Researcher

Birch Scientific, St John's

(2024 April - August)

- Developed equations and algorithms for 3D mesh scale-space filtering using the Laplacian of Gaussian technique, utilizing VTK, Open3D, Trimesh, Gdist, and pygeodesic libraries for multiscale mesh visualization.
- Implemented computationally intensive Scale Space Filtering (SSF) algorithms on NVIDIA CUDA, utilizing parallel programming techniques to significantly enhance processing speed and accuracy.
- Collaborated on a team to build a SaaS data visualization platform with React.js, to generate dynamic graphs, and interact with 3D visualizations via an Unreal Engine plugin.